

Basic Questions:

Q:1 What do you mean by statistics? Discuss the scope and limitations.

Q:2 What is a variable? Discuss different types of variables with examples.

Q:3 What is the scale of measurement? Explain the different types of measurement.

Q:4 Define with examples: i) Population ii) Sample iii) Data

Presentation of data:

Q:5 Imagine we're conducting a survey on the time spent by students studying on weekdays at a university called Blue Ridge University. The university's administration wants to understand the study habits of students to improve academic support services. We surveyed 50 randomly selected students from various departments in Blue Ridge University.

The study times (in minutes) recorded are: 45, 50, 60, 55, 70, 85, 75, 90, 40, 95, 80, 100, 110, 120, 115, 65, 55, 70, 75, 50, 60, 85, 90, 95, 100, 105, 70, 80, 85, 95, 110, 115, 50, 45, 60, 65, 70, 85, 90, 95, 100, 105, 110, 120, 125, 130, 135, 140, 145, 150

- a) Construct an exclusive frequency distribution choosing appropriate class interval.
- b) Using the above data, draw a histogram and a frequency polygon on the same graph.
- c) Determine whether the data is skewed or not from the histogram drawn above.

Q:6 Consider a survey on the weekly exercise duration (in minutes) of residents in a town called Willow Creek. The local health department wants to assess the physical activity levels of residents to promote better health and wellness programs. We surveyed 50 randomly selected residents in Willow Creek.

The exercise durations recorded are: 30, 45, 60, 50, 75, 90, 80, 95, 35, 100, 85, 110, 120, 125, 115, 55, 65, 75, 85, 40, 50, 70, 80, 90, 100, 105, 115, 120, 130, 140, 145, 150, 50, 35, 55, 60, 70, 80, 85, 95, 100, 105, 110, 120, 125, 130, 135, 140, 145, 150

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Measures of Central Tendency:

Q:7 A small company records the number of sales made by each of its 10 sales representatives in a week. The data (number of sales) is as follows: 12, 15, 20, 17, 14, 18, 16, 22, 19, 21.

- a) Calculate the mean, median, and mode of the sales data.
- b) Determine the skewness of the data based on the relationship between mean, median, and mode.
- c) Which of the following central measures is suitable for this data.

Q:8 The following table shows the number of books read by a group of students during summer vacation:

Number of books	Frequency
1	5
2	8
3	10
4	6
5	3

- a) Determine the mean, median, and mode number of books read.
- b) Determine the skewness of the data based on the relationship between mean, median, and mode.
- c) Which of the following central measures is suitable for this data.

Q:9 The weekly income (in dollars) of 50 employees at a factory is grouped as follows:

Income Range (\$)	Frequency
300-400	5
400-500	12
500-600	18
600-700	10
700-800	5

- a) Calculate the mean, median, and mode of the weekly income.
- b) Determine the skewness of the income distribution and comment on its direction.
- c) Which of the following central measures is suitable for this data.

Measures of Location:

Q10: The heights (in cm) of 12 students in a class are recorded as follows: 150, 155, 160, 165, 170, 175, 180, 185, 190, 195, 200, 205.

- a) Determine the first quartile (Q1), median (Q2), and third quartile (Q3) of the height data.
- b) Use quartiles to analyze the skewness of the height data.

Q:11 The ages of participants in a community workshop are shown in the table below:

Age (years)	Frequency
20	6
25	8
30	10
35	5
40	4

- a) Calculate the first quartile (Q1), median (Q2), and third quartile (Q3) of the age distribution.
- b) Use the quartiles to determine the skewness of the distribution.

Q:12 The following table shows the distribution of test scores for 60 students:

Score Range	Frequency
40 - 50	8
50 - 60	12
60 - 70	20
70 - 80	15
80 - 90	5

- a) Find the first quartile (Q1), median (Q2), and third quartile (Q3) of the test scores.
- b) Analyze the skewness of the test scores using the quartiles.

Measures of Dispersion:

Q13: The daily rainfall (in mm) recorded over 7 days in a city is as follows: 12, 15, 18, 20, 25, 30, 35.

Calculate the range, inter-quartile range (IQR) variance, and standard deviation of the rainfall data.

Q:14 The following table shows the number of hours spent on homework by a group of students in a week:

Hours	Frequency
5	4
6	6
7	8
8	5
9	2

a) Determine the range, inter-quartile range (IQR) variance, and standard deviation of the hours spent on homework.

b) Which of the following dispersion measures is suitable for this data.

Q:15 The following data represents the distribution of weights (in kg) of 50 athletes:

Weight Range (kg)	Frequency
50 - 55	7
55 - 60	13
60 - 65	20
65 - 70	8
70 - 75	2

a) Calculate the range, variance, inter-quartile range (IQR) and standard deviation of the athletes' weights.

b) Which of the following dispersion measures is suitable for this data.

Q:16 a) In what situations is the Coefficient of Variation (CV) particularly useful?

b) Football Players' Goals Scored in 10 Matches

- Player A: 2, 3, 5, 4, 2, 6, 3, 7, 4, 5
- Player B: 1, 10, 3, 4, 5, 2, 7, 8, 4, 6

i) Calculate the mean and standard deviation for each player's goals scored over the 10 matches.

ii) Compute the CV for both Player A and Player B.

iii) Based on the CV values, determine which player is more consistent in their performance across the matches.

Q:17 Weekly Closing Prices of Two Shares Over 10 Weeks (in BDT)

- Share X: 150, 155, 152, 160, 158, 162, 165, 167, 170, 172
- Share Y: 200, 180, 220, 250, 210, 190, 230, 240, 260, 250

a) Calculate the mean and standard deviation for the weekly closing prices of both Share X and Share Y.

b) Compute the CV for both shares.

c) Based on the CV values, determine which share is more stable in its price over the 10-week period.

Box and Whisker Plot

Q:18 The blood glucose of 30 females is recorded. The results, in mmol/litre are shown in the stem and leaf diagram below:

Stem	Leaf	Key: 2 1 means 2.1
2	2 2 3 3 5 7	(6)
3	1 2 6 7 7 7 8 8 8 8 9 9 9	(13)
4	0 0 0 4 5 6 7 8	(8)
5	2 5 8	(3)

a) Draw a box plot to represent the data.

The blood glucose level of 30 males is recorded. The results, in mmol/litre, are summarized below:
Lower quartile = 3.6; Upper quartile = 4.7; Median = 4.0; Lowest value = 1.4; Highest value = 5.2

b) draw a box plot for these data on the same diagram as the box plot shown in part (a).

c) compare the blood glucose level for males and females.

Q:19 The runs scored by **Virat Kohli** in 30 One Day International (ODI) matches are shown below using a **stem-and-leaf diagram**:

Stem	Leaf	Key: 2 5 means 25
2	5 9	(2)
3	1 4 7 8	(4)
4	0 1 3 3 5 6 7	(7)
5	0 0 1 1 3 6 6 9	(8)
6	1 5 5 8	(4)
7	0 2 4	(3)
8	0	(1)
9	9	(1)

a) Draw a box plot to represent Kohli's scores.

The runs scored by Babar Azam in 30 ODI matches are summarized below:

Minimum = 22; Lower Quartile (Q1) = 43; Median = 56 ; Upper Quartile (Q3) = 72;
Maximum = 105

b) Draw a box plot for Babar Azam's scores on the same diagram as Virat Kohli's.

c) Compare the performance of Virat Kohli and Babar Azam using the box plots.

Correlation and Regression Analysis

Q:20 In a training scheme for young people, the average time taken for each group to reach a certain level of proficiency was measured. The data are shown below in the table:

Age x (years)	16	17	18	19	20	21	22	23	24	25
Average time y (hours)	12	11	10	9	11	8	9	7	6	8

- Find S_{xx} , S_{yy} and S_{xy} .
- Use your answers in part (a) to calculate the Pearson's Correlation Coefficient (r).
- Describe and interpret the relationship between average time and age.
- Give a reason whether or not the answer in part (b) supports a linear relationship between x and y .

Q:21 Ranpose hospital offers services to a large number of clinics that refer patients to a range of hospitals.

The manager at Ranpose hospital took a random sample of 16 clinics and recorded

- the distance, x km, of the clinic from Ranpose hospital
- the percentage, y %, of the referrals from the clinic who attend Ranpose hospital.

The data are summarised as: $\bar{x} = 8.1$; $\bar{y} = 20.5$; $\sum y^2 = 8266$; $S_{xx} = 368.16$; $S_{xy} = -630.9$

- Find the product moment correlation coefficient for these data.
 - Give an interpretation of your correlation coefficient.
- The manager at Ranpose hospital believes that there may be a linear relationship between the distance of a clinic from the hospital and the percentage of the referrals who attend the hospital.
- Find the equation of the regression line of y on x , giving your answer in the form $y = a + bx$
 - Give an interpretation of the gradient of your regression line.

Q:22 The chairman of a marketing department at a large private university undertakes a study to relate starting salary (y) in thousands after graduation for marketing majors to grade point average (GPA) in major courses. To do this, records of 10 recent marketing graduates are randomly selected. The GPA (x) and the corresponding starting salary were as follows:

GPA (x)	3.26	2.60	3.35	2.86	3.82	2.21	3.47	3.28	2.54	3.25
Salary (y) in thousands	33.8	29.8	33.5	30.4	36.4	27.6	35.3	35.0	26.5	33.8

- Find the equation of the least squares regression line of y on x .
- Give an interpretation of the value of b in your regression line.
- Find the point prediction of starting salary corresponding to the GPA 3.75.
- Comment on the reliability of your answer in part (b). Give a reason, in context, for your answer.

Q:23 Martin is investigating the relationship between a person's daily caffeine consumption, c milligrams, and the amount of sleep they get, h hours, per night. He collected this information from 20 people and the results are summarized below.

$\Sigma c = 3660$; $\Sigma h = 126$; $\Sigma c^2 = 973228$; $S_{hh} = 43.7349$; $\Sigma ch = 20023.4$; $S_{cc} = 303448$; $S_{ch} = -3034.6$.

- Calculate the product moment correlation coefficient between c and h .
- Give a reason whether or not the answer in part (a) supports a linear relationship between c and h .
- The amount of sleep per night is the response variable. Explain what you understand by the term 'response variable'.
- Martin says that for each additional 100 mg of caffeine consumed, the expected number of hours of sleep decreases by 1. Determine, by calculation, whether or not the data support this statement.
- Use the data to calculate an estimate for the expected number of hours of sleep per night when no caffeine is consumed.

Q:24 Stuart is investigating the relationship between Gross Domestic Product (GDP) and the size of the population for a particular country. He takes a random sample of 9 years and records the size of the population, t millions, and the GDP, g billion dollars for each of these years.

The data are summarized as $n=9$; $\Sigma t = 7.87$; $\Sigma g = 144.84$; $\Sigma g^2 = 3624.41$; $S_{tt} = 1.29$; $S_{tg} = 40.25$.

- Stuart calculates the product moment correlation coefficient for these data and obtains 0.985. Give an interpretation of the product moment correlation coefficient.
- Find the equation of the least squares regression line of g on t in the form $g = a + bt$.
- Give an interpretation of the value of b in your regression line.
- Use the regression line from part (b) to estimate the GDP, in billions of dollars, for a population of 7,000,000.
- Comment on the reliability of your answer in part (d). Give a reason, in context, for your answer.

Probability

Q:25 a) Find the number of arrangements of the letters of the word **STATISTICS** using all the letters in each arrangement.

b) State the important properties of Normal distribution. Name a random variable, which has a normal distribution.

c) The random variable $X \sim N(\mu, \sigma^2)$. Given that $P(X < 17) = 0.8159$ and $P(X < 25) = 0.9970$, find the value of μ and the value of σ .

d) The random variable $X \sim N(50, 4^2)$. Find

i) $P(X < 53)$ ii) $P(X > 58)$

Q:26 A psychologist gives a student two different tests. The first test has a mean of 80 and a standard deviation of 10 and the student scored 85.

a) Find the probability of scoring 85 or more on the first test.

The second test has a mean of 100 and a standard deviation of 15. The student scored 105 on the second test.

b) Find the probability of a score of 105 or more on the second test.

c) State, giving a reason, which of the student's two test scores was better.